HENRY KOFI BRANDOH

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RESEARCH INTERESTS

Medical AI, computer vision for clinical data, trustworthy AI for low-resource healthcare settings.

EDUCATION

Ashesi University **Berekuso, Eastern Region, Ghana**
MPhil Intelligent Computing Systems **Expected Sep 2028**
Thesis: Predicting pulmonary embolism risk in asymptomatic individuals (Supervisor: Prof. Martin Nartey-Tamatey)

Ghana Communication Technology University **Accra, Greater Accra, Ghana**
BSc. Information Communication Technology **Aug 2020**
Cumulative GPA: 3.4/4.0

PROJECTS AND RESEARCH

ECG-Vision: Atrial Fibrillation from ECG Images – Independent Project (github.com/henryhkb/ecg-vision-app) **Oct 2026**
- Fine-tuned a ResNet-18 (PyTorch) to detect atrial fibrillation from 12-lead ECG images, trained on PTB-XL with patient-separated splits; test F1 0.954 (sensitivity 0.96, specificity 0.99) on 967 held-out ECGs.
- Found the first model collapsed under photo-like distortions (F1 0.27–0.52 for blur, tilt, perspective, lighting); retrained with augmentation, raising test F1 to 0.92–0.96 while keeping clean performance.
- Validated on 36 real phone photos of ECGs (F1 0.94) and through a blind cardiologist review of the model's errors, which agreed with the model on 6 of 7 cases, exposing label noise in the dataset.
- Deployed a public Gradio demo on Hugging Face Spaces: huggingface.co/spaces/Henryhkb/ecg-vision

Pulmonary Embolism Risk Prediction – MPhil Thesis, Ashesi University **Sep 2026 - Present**
- Developing a model to predict future pulmonary embolism risk in currently asymptomatic people from lifestyle and clinical risk factors, piloting on PhysioNet data before collecting data from Ghanaian hospitals.

SafeArrive: Maternal Risk Triage App – Developer, Harvard HSIL Hackathon **Apr 2026**
- Created a progressive web app (React/Vite, FastAPI) with an XGBoost risk classifier trained on the Kaggle Maternal Health Risk dataset: 71% overall accuracy and 81% precision on high-risk cases.
- Added a Twi/English interface and Arkesel SMS referrals so Community Health Workers can use it without internet access.

MetaCalib: Metacognitive Benchmark for LLMs – Kaggle × Google DeepMind Hackathon **Mar 2026**
- Independently designed a four-task metacognition benchmark for LLMs (calibration, difficulty prediction, knowledge limits, self-correction); scored 81.25% (13/16) on Kaggle agent exams.

Twi Language Model – NLP Coursework, Ashesi University **Sep 2026**
- Building n-gram language models for Twi (Akan), a low-resource Ghanaian language, and adapting them to limited training data.

WORK EXPERIENCE

Orpasser – Founder & Lead Developer, Accra, Ghana **Jan 2022 - Present**
- Cut organisers' trip setup from days to under 10 minutes by developing a full-stack booking platform (Laravel, MySQL) with online bus booking, QR tickets and SMS alerts.
- Registered the company (2025) and piloted it with organisers, finding rental quotes near double their budgets (GH₵13,000 vs GH₵7,000), which reshaped the plan around owning buses.
- Engineered a 3-party escrow payment flow holding 100% of passenger payments until trip completion, then paying organisers via Mobile Money, tracked in an admin dashboard.

National Cardiothoracic Centre, Korle Bu Teaching Hospital – IT Manager, Accra **Jun 2022 - Aug 2026**
- Supervised and mentored a team of 5 (3 interns, 2 national service personnel), delivering IT support to 50+ doctors, nurses and clinical staff.
- Performed daily health checks on the centre's main server rack, data-centre link and departmental devices, keeping the Hospital Management System available to 50+ clinical staff.
- Implemented a letter and memo tracking system adopted by secretaries across about 10 hospital departments, replacing paper logbooks and enabling date-range reporting.

Freelance – Full-Stack Developer, Accra, Ghana **Jan 2020 - Present**
- Delivered a live e-commerce store for a paying fashion-boutique client (Laravel, Bootstrap 5, MySQL), with Paystack online payments, guest checkout and an admin panel for products and orders.

CERTIFICATIONS

CITI Program: Data or Specimens Only Research (in progress) • PhysioNet Credentialed Researcher (in progress)

CO-CURRICULAR ACTIVITIES

Blood Cancer Foundation Ghana – Volunteer Web Manager **Apr 2026 - Present**
- Manage and maintain the foundation's website, keeping campaign, event and awareness content up to date for the public.

SKILLS

- Languages: Python, PHP, JavaScript / TypeScript, MySQL, Bash
- AI / ML: PyTorch (CNN transfer learning, data augmentation), XGBoost, scikit-learn, LangChain, LLM APIs (Claude, GPT-4, LLaMA via Groq, Gemini), model evaluation and threshold selection; TensorFlow (learning)
- ML Tools: Google Colab (GPU), Gradio, Hugging Face Spaces, Git / GitHub
- Frameworks: Laravel, React, FastAPI, Node.js, Bootstrap, Flutter (basic)
- APIs & Tools: REST APIs, Postman, Paystack, Mobile Money, Arkesel SMS
- Soft Skills: Leadership, Mentoring, Problem-Solving, Stakeholder Communication